

Obstacles is one of the main problems in order to control omnidirectional mobile robot. Most mobile robots use simple algorithm to avoid obstacles that allows the robot identify these obstacles. To detect obstacles, the most important factor is to define the distance from the obstacles to stereo cameras to determine the depth of field. Most commercial stereo cameras are relatively high cost and they can only determine the distance from the far. They limit the flexibility and performance of robot that these cameras do not let robot define the obstacles at close range. At the same time, the development of image processing system is based on embedded system which is a difficult problem that need to search in further research.

(43)
This study is based on embedded system for self-propelled robot in order to develop image processing applications that help the robot determine the direction and avoid obstacles, implementing the translational status flexibility including turn left, turn right, go straight ahead and stop. The robot control system is programmed and applied on STM32F4-Discovery module. Three-wheeled robot operation based primarily on information and control commands from the block sends the image processing. The control block performs 'avoid obstacles' based on the information and the relative distance. It was found that this tricycle omnidirectional robot recognized objects within 20cm - 80cm. In this distance the robot works on image processing system combined with encoder rotation on three engines to determine the distance and angle of the robot reality. The error in the distance from the stereo cameras to the object is about 1 cm which is an acceptable level for robot operation based on image processing system. However, there exist some drawbacks to be overcome, such as image processing system was slow, the robot sometimes could not follow the completely controlled trajectory and the operation of the robot model is limited.

Q5. Read the following case study. Write an informal report using the headings:

20 marks CLO 07

- | | |
|--------------------|---------|
| 1) Title | 2 marks |
| 2) Introduction | 4 |
| 3) Methodology | 4 |
| 4) Findings | 4 |
| 5) Conclusion | 4 |
| 6) Recommendations | 2 |

(1) The Internet of Things (IoT) is a recent communication paradigm that envisions a near future, in which the objects of everyday life will be equipped with microcontrollers, transceivers for digital communication, and suitable protocol stacks that will make them able to communicate with one another and with the users, becoming an integral part of the Internet. The IoT concept, hence, aims at making the Internet even more immersive and pervasive. Furthermore, by enabling easy access and interaction with a wide variety of devices such as, for instance, home appliances, surveillance cameras, monitoring sensors, actuators, displays, vehicles, and so on, the IoT will foster the development of a number of applications that make use of the potentially enormous amount and variety of data generated by such objects to provide new services to citizens, companies, and public administrations. This paradigm indeed finds application in many different domains, such as home automation, industrial automation, medical aids, mobile healthcare, elderly assistance, intelligent energy management and smart grids, automotive, traffic management, and many others.

However, such a heterogeneous field of application makes the identification of solutions capable of satisfying the requirements of all possible application scenarios a formidable challenge. This difficulty has led to the proliferation of different and, sometimes, incompatible proposals for the practical realization of IoT systems. Therefore, from a system perspective, the realization of an IoT network, together with the

Q2. Write a cover letter in response to the following job advertisement.

Job Description:

We are looking for an innovative electronics/ electrical engineers for day shifts at our Sheikhpura industry. The candidates should have the passion to turn ideas into realities. Are you passionate about building sensors and electronics from the ground up? Are you a talented individual who understands electronics more than most people do? The ideal candidate has outstanding engineering skills, passion to tackle technically challenging areas and a deep desire to create breakthrough technology. We are looking for skilled and motivated individuals for electronics hardware development and product development. These individuals will be working alongside teams of other engineers in the front lines of product development in a fast-paced, dynamic environment.

Responsibilities:

- Design and development of electronic sensor and components
- Collaborative work with app developers and other engineers
- Start to finish development from specification to design, prototype, and manufacturing.
- Product level electrical design including component selection

Requirements:

- BS or higher degree in Electrical Engineering
- Relevant coursework or work experience (*Fresh candidates are equally eligible*)
- Experience in electrical engineering within a product design and manufacturing environment
- Experience in selection and use of DSPs, microcontrollers, and programmable logic devices and their peripheral circuitry (signal conditioning, board-level power supplies)
- Experience in design, component selection, and PCB implementation of analog signal conditioning circuitry for input and output signals within digitally-controlled electronics systems
- Experience in development of digital and analog interfaces for PC-based test systems, especially (Labview, Matlab, Python)
- Experience in supervision of PCB layout, plane definition, and EMC optimization

Apply to the Manager HR through regular postal mail before May 24th, 2017, at the following address: XYZBLINKS, 25 km, Sheikhpura.

Q3. You are applying for higher education abroad in the field of your choice. Write a research statement stating reasons for applying in this university, summary of your research accomplishments, current work, future direction and potential of your work.

15 marks CLO 05

About me - Education - Research -

Q4. Write an **Abstract** for the following case. Use 100 – 125 words and circle the number of words counted. Label the elements of your abstract. Write 3 – 4 key words as well. 10 marks CLO 06, 07

Topic: An application of movement direction control for the three wheeled mobile robots using visual information.

Nowadays, image processing and computer vision are being strongly developed and largely applied in many different fields in our lives. The application of image processing and object recognition are being developed in many fields such as license plate recognition system, fingerprint recognition system, and automatic fire alarm system. These systems are being developed in order to achieve their optimization in the future. This study aims to make self-propelled robots to replace how humans do housework in the home or even in polluted environments. Avoidance.

National University of Computer and Emerging Sciences, Lahore Campus



Course Name:	Technical Report Writing (TRW)	Course Code:	SS-142
Program:	EE	Semester:	Spring 2017
Duration:	180 Minutes	Total Marks:	80
Paper Date:	19 th May, 2017	Weight:	50%
Section:	ALL	Page(s):	4
Exam Type:	Final		

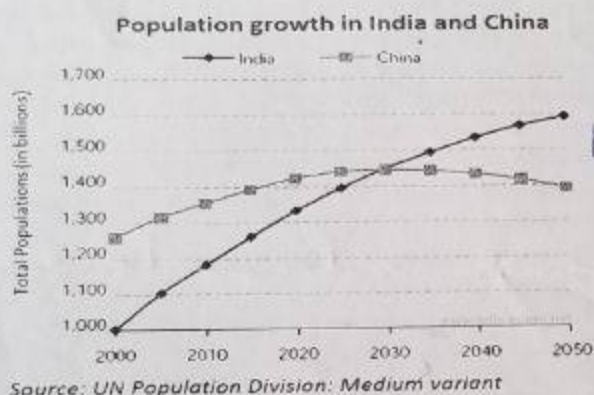
Ayesha Orfan

LI4-4405

Instructions/
Notes:

1. DO NOT solve on the Question paper. Write only in the given answer copy.
2. Untidy work/ cutting/ erasing/ overwriting would lose marks.
3. Proof-read your work very carefully before submission.

Q1. (a) The graph below shows population figures for India and China since the year 2000 and predicted population growth up until 2050. Summarize the information by selecting and reporting the main features, and make comparisons where relevant. (Limit 120 – 150 words). 10 marks CLO 03, 04



24-04

19-07-09 → 50

43

(b) Paraphrase the following passage:

5 marks CLO 02

Fog computing has been introduced as a technology to bridge the gap between remote data centers and Internet of Things (IoT) devices. Enabling a wide range of benefits, including enhanced security, decreased bandwidth, and reduced latency, fog is an appropriate paradigm for many IoT services. Nevertheless, fog devices (located at the edge of the Internet) obviously face many security and privacy threats. Generally speaking, the goals of the fog computing paradigm are to reduce the data volume and traffic to cloud servers, decrease latency, and improve quality of service (QoS). *IEEE Internet Computing (Volume: 21, Issue: 2, Mar.-Apr. 2017)*

Recognition
propelled — moved
Module — Layout of system
Obstacles — hindrances.

required backend network services and devices, still lacks an established best practice because of its novelty and complexity. In addition to the technical difficulties, the adoption of the IoT paradigm is also hindered by the lack of a clear and widely accepted business model that can attract investments to promote the deployment of these technologies.

Method

In this complex scenario, the application of the IoT paradigm is to implement in cities of particular interest, as it helps governments to adopt ICT solutions in the management of public affairs, thus realizing the so-called Smart City concept whose final aim is to make a better use of the public resources, thus increasing the quality of the services offered to the citizens, while reducing the operational costs of the public administrations. This objective can be pursued by the deployment of an urban IoT, i.e., a communication infrastructure that provides unified, simple, and economical access to a plethora of public services, thus unleashing potential synergies and increasing transparency to the citizens. The objective of this paper is to discuss a reference framework for the design of an urban IoT by implementing this project in the "Pedova Smart City" project, which is a proof-of-concept deployment of an IoT island in the city of Pedova (Italy).

The project assisted the public in noise monitoring, traffic congestion, waste management, smart lighting, smart parking, and so on. These services are based on a centralized architecture, where a dense and heterogeneous set of peripheral devices deployed over the urban area generate different types of data that are then delivered through suitable communication technologies to a control center, where data storage and processing are performed. The data collected by the urban IoT is easily accessible by authorities and citizens, to increase the responsiveness of authorities to city problems, and to promote the awareness and the participation of citizens in public matters.

There is deployment of suitable protocol layers in the different elements of the network. Then link layer technologies are used to interconnect the different parts of the IoT. The focus is specifically on IETF standards because they are open, free and are based on Internet best practices, and can serve a wide community. The data exchange is typically accompanied by using eXtensible Markup Language (XML). The discussed technologies are close to being standardized, and industry players are already active in the production of devices that take advantage of these technologies to enable the applications of interest. In fact, while the range of design options for IoT systems is rather wide, the set of open and standardized protocols is significantly smaller. The enabling technologies, furthermore, have reached a level of maturity that allows for the practical realization of IoT solutions and services, starting from field trials that will hopefully help clear the uncertainty that still prevents a massive adoption of the IoT paradigm.

[Text adapted from: IEEE Internet of Things Journal (Volume: 1, Issue: 1, Feb. 2014) p. 22 - 32]

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Technical Report Writing	Course Code:	SS - 142
Program:	BS (Electrical Engineering)	Semester:	Spring 2018
Duration:	60 Minutes	Total Marks:	35
Paper Date:	12 th April, 2018	Weight	15%
Sections:	All	Pages:	1

Instruction/ Notes: DO NOT solve on the question paper. Write only in the given answer copy.
Avoid cutting/ erasing/ overwriting.

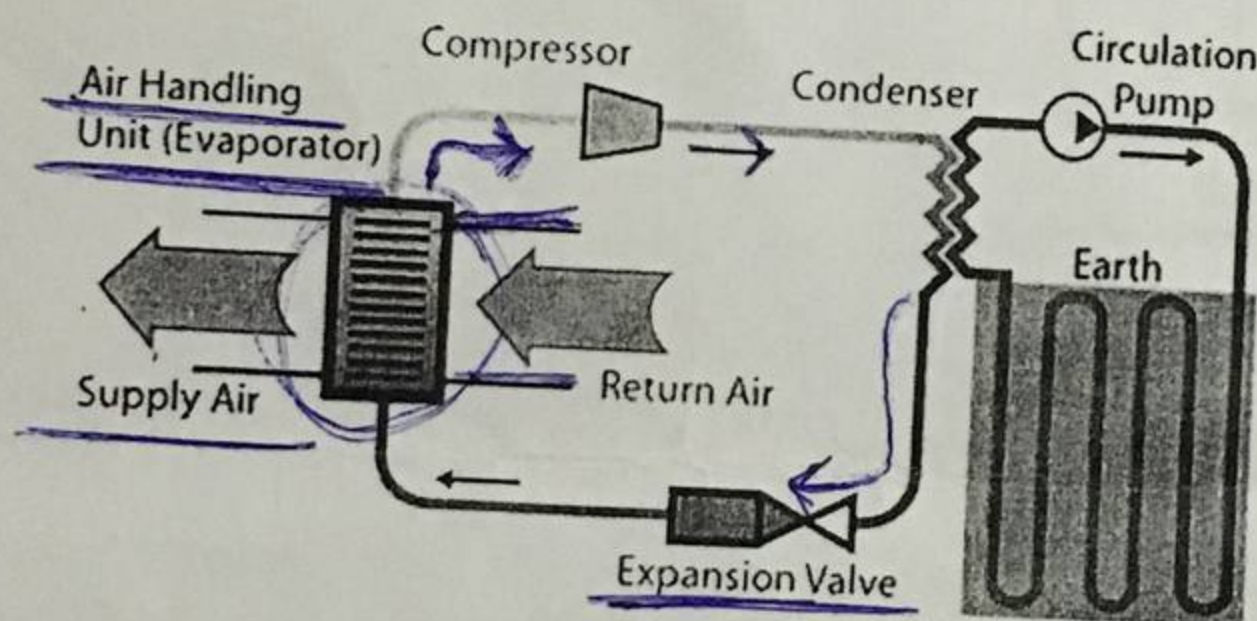
Q 1. Paraphrase the following passage:

(CLO - 06) (10 marks)

Cloud-based services, social networks, and multimedia streaming have rapidly increased the demand for high-bandwidth mobile communications. In addition, consumers also expect long battery life from mobile devices. One solution may be a low-energy amplifier prototype developed by European researchers to prolong battery life for 5G communication. 5G wireless broadband will provide bandwidth of multiple gigabits per second, requiring wider bands at higher frequency ranges than today's communication systems use. (ieeexplore, 2018).

Q 2. Describe the following figure in a 75 - 100 words cohesive paragraph. Also write its title.

(CLO - 03) (10 marks)



Q 3. As an electrical engineer at Green Automations Ltd., you are responsible for electrical operations and maintenance activities. You also supervise electrical and electronic technicians and junior engineers who send you daily activity report of all maintenance and fault analysis. Recently, they pointed out in their reports that the equipment at semi-automatic and manual cable routing is causing problems which is leading to other problems at different workstations. Write an informal **Equipment Evaluation memo report** to your Chief Engineer stating:

Date:

To:

From:

Subject:

1. Purpose of writing
2. The equipment's normal working patterns
3. Routing problem encountered
4. Supportive evidence
5. Conclusion/ possible solution(s)

Be precise. Follow guidelines for technical writing.

(CLO - 05) (15 marks)

Name: Ayesha JawedRoll No: L15-4522

Instruction/ Notes:

No answer sheets required.

Q. The following memorandum does not meet the requirements of a good technical document.

Rewrite it to establish its technical clarity.

15 marks. (CLO – 01, 02)

(Adapted from: Writing that works/Gerson)

Date: February 9, 2018

To: Zaheer Ehsan

(From: *(Your name)*)

Subject: exhibition

(Please prepare to participate in expo 18

Make sure the information is very detailed.

Thanks.

Date: February 9, 2018

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Technical Report Writing	Course Code:	SS - 142
Program:	BS (Electrical Engineering)	Semester:	Spring 2018
Duration:	180 Minutes	Total Marks:	100
Paper Date:	14 th May. 2018	Weight	50%
Exam:	End Term	Pages:	4
Sections:	All		

Instruction/ Notes: DO NOT solve on the question paper. Write only in the given answer copy.
Avoid cutting/ erasing/ overwriting.

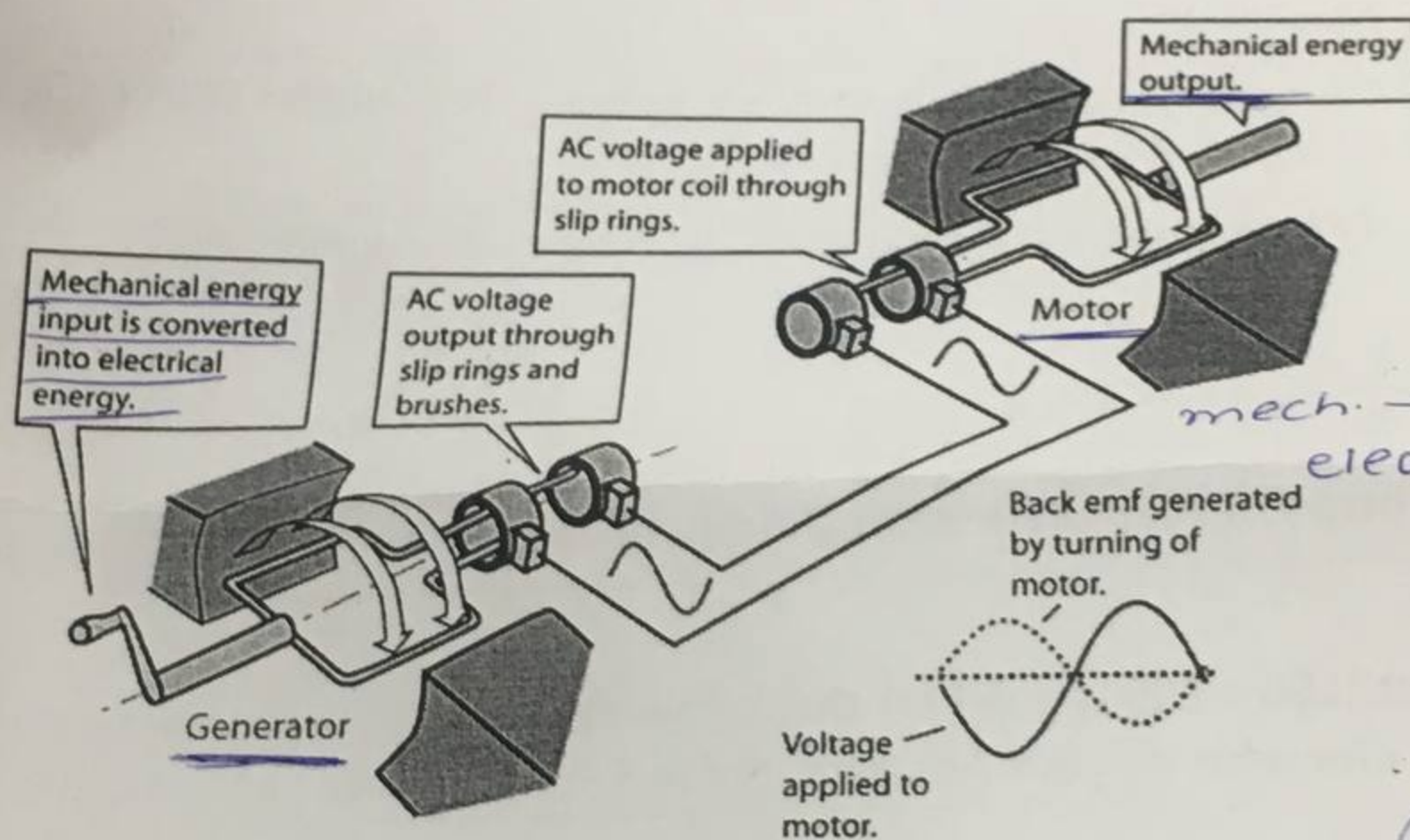
L15-4522

Q1. Describe the schematic figure given below in 150 – 200 words. Your description should include the following:

- A Title to this diagram
- Its working theory/ principle
- Expected behaviour with 2 different voltages
- Its 3 applications in the real life/ industry

CLO - 03, 04

(10 marks)



Q2. (a) Paraphrase the following passage:

CLO 02

(10 marks)

Deep learning has in recent years set an exciting new trend in machine learning. The theoretical foundations of deep learning are well rooted in the classical neural network (NN) literature. But different to more traditional use of NNs, deep learning accounts for the use of many hidden neurons and layers—typically more than two—as an architectural advantage combined with new training paradigms. While resorting to many neurons allows an extensive coverage of the raw data at hand, the layer-by-layer pipeline of nonlinear combination of their outputs generates a lower dimensional projection of the input space. Every lower-dimensional projection corresponds to a higher perceptual level. Provided that the network is optimally weighted, it results in an effective high-level abstraction of the raw data or images. This high level of abstraction renders an automatic feature set, which otherwise would have required hand-crafted results or bespoke features. (IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS, VOL. 21, NO. 1, JANUARY 2017)

(b) Following IEEE referencing guidelines, write end references from information given below:

CLO – 08

(5+5=10 marks)

1	
Title	Deep Learning for Health Informatics
Author	D. Ravi et al.
Date of Publication	January 2017
Journal	IEEE Journal of Biomedical and Health Informatics
DOI	10.1109/JBHI.2016.2636665
Printed Page No	4-21
Issue No	1
Volume No	21

2	
Authors	S. Jeong, Y. J. Jang, D. Kum and M. S. Lee
Date of Publishing	May 2018
Title	Charging Automation for Electric Vehicles: Is a Smaller Battery Good for the Wireless Charging Electric Vehicles?
Title of Journal	IEEE Transactions on Automation Science and Engineering
Vol. number	1
Issue	3
URL	http://ieeexplore.ieee.org/tamp.jsp?tp=&arnumber=8353730&is=435806

Q3. Write an Abstract (150 – 200 words) for the following case. Circle the number of words counted and label the elements of your Abstract. Write 3 – 5 key words as well. CLO – 6 (15 marks)

Topic: Charging Automation for Electric Vehicles: Is a Smaller Battery Good for the Wireless Charging Electric Vehicles?

Interest in electric vehicles (EVs) has recently grown due to calls for ecofriendly transportation. Battery-powered electric buses or plug-in EV buses, which produce zero tailpipe emissions, offer significant potential in improving sustainability and an ecofriendly environment in urban areas. EV-based transit buses require a large battery for a long service time. For instance, a long-range all-electric bus manufactured by BYD Auto Company has a 324-kWh lithium iron phosphate (LFP) battery [1]. Unfortunately, the large capacity of the batteries of current EV buses prevents them from gaining popularity as a mainstream mass transit solution. The current problems of plug-in EV buses are the long operational idle during the battery charging time, the high cost of the battery, and the great weight of the battery [2]. Dynamic wireless charging (DWC) systems have emerged as an alternative to address the challenges caused by the current battery technology. DWC has been successfully applied in the charging of handheld devices, such as electronic toothbrushes, phones, and medical devices. It has also been

widely used for automated material handling systems in semiconductor fabrication and LCD production lines [3], [4]. Recently, the technology has been applied to EVs. A DWC system is a road-integrated EV system that is composed of a roadway wireless charger called a power track and EVs. The EVs receive electric power from the power track without any contact, even while in motion [5]. That is, when an EV passes over the pavement under which the power track is installed, an electromagnetic field is produced from the power track (power transmitter) upward to the EV. A pickup device (power receiver) at the bottom of the EV receives the electromagnetic field and converts it into electric power to operate the traction motor or charge the battery [6], [7]. We adopt a known theory on capacity degradation of the battery through its lifetime. That is, the capacity of the battery is no longer considered to be a static variable as in the previous DWC vehicle system model in [9] and [16]–[18]. Rather, we treat the battery capacity as a dynamic variable that decreases over the battery's lifespan. A well-known lifetime capacity prediction model called Ah-throughput model is used in the analysis. Our quantitative model indicates that the DWC can positively improve battery lifetime, but this positive effect is always true only when the battery is large enough, even if the battery is charged frequently with DWC. Through the results, we confirm that the DWC system is considerably beneficial for battery life and that those designed with the proposed optimization model can be economically robust.

Q 4. Write a cover letter in reference to the following job posting.

CLO - 02 (20 marks)

ELECTRICAL ENGINEERS REQUIRED

Job Description: ElectroFux® is looking for young electronics/ electrical engineers for their electronic components manufacturing unit at Gujrat. The candidates should have the passion to turn ideas into realities of the LEDs and small consumer electronic items. These individuals will be working alongside teams of other engineers in the front lines of product development in a fast-paced, dynamic environment.

Responsibilities:

- Design and development of electronic sensors, LEDs and components
- Collaborative work with app developers and other engineers

Requirements:

- BS or higher degree in Electrical Engineering
- Relevant coursework or work experience (Fresh candidates are equally eligible)
- Experience in supervision of PCB layout, plane definition, and EMC optimization

Apply to the Manager HR through regular postal mail before May 30th, 2018, at the following address:
The Manager HR, ElectroFux®, 10 Km, Ferozepur Road, Lahore.

Q5. Read the following case study. Write an informal report using the headings: CLO - 8 (20 marks)

- | | |
|--------------------|---------|
| 1) Title | 2 marks |
| 2) Introduction | 4 |
| 3) Methodology | 4 |
| 4) Findings | 4 |
| 5) Conclusion | 4 |
| 6) Recommendations | 2 |

Date:
To
From
P.S.

S.P.K.S

Quality assessment during post-construction of buildings is an indispensable procedure in construction industry. This paper describes the design and development of a quality inspection and assessment robot (QuicaBot) that can autonomously scan the entire room using cameras and laser scanners to pick up building defects, such as hollowness, crack, evenness, alignments, and inclination. A robotic system consisting of four types of sensors and a mobile platform as well as the corresponding five types of assessment algorithms is proposed. To the best of our knowledge, this paper is the first attempt to have a complete robotic system for post-construction quality assessment of buildings. The aim of the developed system is twofold: first, to systematize the manual inspection work through automation resulting in more reliable and objective inspection reports, and then, to speed up the inspection process resulting in a more efficient end product. Based on our experimental on-site tests, the developed novel robot takes only half of the manual inspection time when inspecting the same room. We have also observed that the autonomous assessment results have better inspection accuracy when compared to manual assessments. Last but not least, the results provided by QuicaBot have more consistent measurement accuracy when compared to a manual assessor. Motivated by the initial successful on-site tests and as being a practical mechatronic system illustrating how sensing, sensor fusion, and actuation can be integrated to achieve an intelligent system for building defects assessment, we believe that the QuicaBot-like robots are going to become an integral part of construction industry in the near future.

Post construction quality assessment of buildings is an indispensable procedure in construction industry which is currently executed by the manual inspectors. According to the building and construction authority (BCA) of Singapore [1], which is responsible for manual assessment of all the buildings in Singapore, in a standard daily quality inspection operation, at least two inspectors, one for using the manual assessment tools and the other is for preparing reports, are needed to complete the manual operation. Such a manual assessment procedure may import several errors into inspection because of executing the operation in incorrect way or the use of inaccurate inspection tools. What is more, the manual inspector may get tired after some time and the inspection accuracy may decrease over time. Last but not least, in most of the time, manual inspection has to be done during daytime. Considering this time-consuming, tiresome, and unexciting procedure [1], and motivated by huge market requirement of construction robots referring to UNESCO Science Report [2] and World Economic Forum, we propose an automated post-construction quality assessment robotic system consisting of a mobile robot, a 2-D laser scanner, a color camera, a thermal camera with heater, and an inclinometer. Although there are many other defects, which are being inspected by manual inspectors in today's technology, we consider only five defects for the designed robot in this paper: hollowness, crack, evenness, alignment, and inclination. The prototype is tested both in a constructed test bed and a CONQUAS room to validate the robustness and accuracy of the proposed methods. All these autonomous assessments have a close even better inspection accuracy with the ones in manual assessments. When compared to manual assessment, the proposed assessment system has higher effectiveness and more consistent measurement accuracy. Moreover, a few of these robots working together will make building inspection better, as they can feed data back to the human operator to inspect critical and complex defects that are not considered by the robot. (IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, May 2018)

Q6. To pursue higher education, you are applying for Masters in Electrical Engineering program at Purdue University. They have asked you to write a Personal Statement in 350-400 words in order to complete your admission-application process. In this statement, describe why you are taking up a specialized area, how you can relate yourself to this program and what engineering problem(s), of personal or national importance, you'd like to solve during and after this course. Also explain how you came to decide about this University's program to achieve your goals? CLO - 07 (15 marks)